DATA CLEANING - PARSING DATES

Check the data type of our date column

```
import pandas as pd
import numpy as np
import seaborn as sns
import datetime

# read in our data
landslides = pd.read_csv("C:\\datasets_686_1296_catalog.csv")

print (landslides.head())
# print the first few rows of the date column
print(landslides['date'].head())

# check the data type of our date column
print (landslides['date'].dtype)
print (landslides['id'].dtype)
print (landslides['continent_code'].dtype)
print (landslides['distance'].dtype)
```

Convert our date columns to datetime

The basic idea is that you need to point out which parts of the date are where and what punctuation is between them. There are lots of possible parts of a date, but the most common are %d for day, %m for month, %y for a two-digit year and %Y for a four digit year.

Some examples:

- 1/17/07 has the format "%m/%d/%y"
- 17-1-2007 has the format "%d-%m-%Y"

```
# create a new column, date_parsed, with the parsed dates
date_parsed = pd.to_datetime(landslides['date'], format="%m/%d/%Y")

print (date_parsed.shape)
print (date_parsed.dtype)

# get the day of the month from the date_parsed column
day_of_month = date_parsed.dt.day
print (day_of_month.head())
```

- **What if I run into an error with multiple date formats?**

  While we're specifying the date format here, sometimes you'll run into an error when there are multiple date formats in a single column. If that happens, you have pandas try to infer what the right date format should be. You can do that like so:

```
landslides['date_parsed'] = pd.to_datetime(landslides['Date'], infer_datetime_format=True)
```

- **Why don't you always use infer_datetime_format = True?**

  There are two big reasons not to always have pandas guess the time format. The first is that pandas won't always been able to figure out the correct date format, especially if someone has gotten creative with data entry. The second is that it's much slower than specifying the exact format of the dates.